Attention Entropy Profiling for Training-Free Out-of-Distribution Detection in Vision Transformers

Anonymous Author(s)

Abstract

We propose Attention Entropy Profiling (AEP), a training-free method that detects out-of-distribution (OOD) inputs to Vision Transformers (ViTs) by analyzing layerwise self-attention statistics. AEP constructs a compact profile vector from four per-layer attention statistics—CLS attention entropy, average token entropy, attention concentration ratio, and head agreement—and scores test inputs by their Mahalanobis distance from in-distribution reference statistics computed from a small calibration set. Unlike existing post-hoc OOD detectors that operate on output logits or penultimate-layer features, AEP captures the model's internal processing dynamics through attention structure, providing a complementary signal. Across experiments with three ViT architectures (DeiT-Small, DeiT-Base, ViT-Base/16) evaluated on five OOD datasets using Food-101 as the in-distribution proxy, AEP achieves AUROC scores of 0.92–1.00, and fusing AEP with feature-based detectors yields the best overall performance. Statistical tests confirm that entropy profiles differ highly significantly between in-distribution and OOD inputs ($p < 10^{-16}$, Hotelling's T^2). Ablation studies reveal that CLS attention entropy is the most informative component and that performance stabilizes with as few as 250 calibration samples. We note important limitations of our evaluation setup—particularly the use of a non-standard ID dataset on which models have zero classification accuracy—and discuss their implications for interpreting results.

1 Introduction

Vision Transformers (ViTs) [Dosovitskiy et al., 2021] have become the dominant architecture for image classification. Deploying ViTs in safety-critical applications—autonomous driving, medical imaging, industrial inspection—requires reliable uncertainty estimation: the model must signal when its predictions are unreliable. A key component of this is *out-of-distribution (OOD) detection*: identifying inputs that differ substantially from the training distribution.

Current post-hoc OOD detection methods for ViTs predominantly rely on output-level signals: maximum softmax probability (MSP) [Hendrycks and Gimpel, 2017], energy scores [Liu et al., 2020], or penultimate-layer features (Mahalanobis distance [Lee et al., 2018], ViM [Wang et al., 2022], KNN [Sun et al., 2022]). These methods treat the transformer as a black box, ignoring the rich intermediate representations computed during inference. In particular, they overlook the *self-attention maps*, which encode how the model distributes computational focus across spatial tokens at each layer.

We observe that the *self-attention entropy profile*—a vector capturing how attention entropy varies across layers and heads—serves as a direct, interpretable signature of how a ViT processes an input. For in-distribution images, the model follows a characteristic processing trajectory: early layers attend broadly (high entropy), while later layers progressively focus on discriminative regions (decreasing entropy). Under distribution shift, this trajectory is disrupted: the model either fails to focus or focuses on irrelevant features.

Preprint. Under review.

Based on this observation, we propose **Attention Entropy Profiling (AEP)**, a training-free, post-hoc method that extracts OOD detection scores from the self-attention maps of any pretrained ViT. AEP requires only a small in-distribution calibration set ($\sim$1,000 images) to compute reference statistics and leverages attention maps that are already computed during the forward pass.

Our contributions are:

- We propose AEP, the first systematic use of layerwise attention entropy profiles for OOD detection in Vision Transformers. AEP is training-free, requires no architectural modifications, and works with any ViT variant.
- We demonstrate that AEP achieves competitive OOD detection performance (AUROC 0.92–1.00) and that fusing AEP with feature-based detectors consistently outperforms either approach alone.
- We provide comprehensive ablation studies showing that CLS attention entropy is the most informative component, late layers are most discriminative, and performance is robust to calibration set size (stable with $\geq$250 samples).
- We provide statistical evidence that attention entropy profiles differ highly significantly between in-distribution and OOD inputs ($p < 10^{-16}$), validating the core hypothesis.

Important caveat. Our experiments use Food-101 as the in-distribution dataset with ImageNet-pretrained models that have zero classification accuracy on Food-101. While this setup validates that attention entropy profiles encode distributional information detectable without task-specific fine-tuning, it constitutes a non-standard evaluation that limits comparison with published benchmarks. We discuss this limitation in detail in Section 5.

The rest of the paper is organized as follows: Section 2 reviews related work, Section 3 describes the AEP method, Section 4 presents experimental results and ablations, Section 5 discusses limitations, and Section 6 concludes.

2 Related Work

Post-hoc OOD detection. The dominant paradigm for OOD detection uses post-hoc scoring on pretrained classifiers. MSP [Hendrycks and Gimpel, 2017] uses the maximum softmax probability. Energy Score [Liu et al., 2020] uses the log-sum-exp of logits. Mahalanobis Distance [Lee et al., 2018] measures feature-space distance from class-conditional Gaussians. ViM [Wang et al., 2022] projects features onto a virtual logit dimension. KNN [Sun et al., 2022] uses k-nearest neighbor distances in feature space. ASH [Djurisic et al., 2023] prunes and scales activations. All of these methods operate on output logits or penultimate-layer features. None leverage the self-attention maps unique to transformer architectures. AEP is the first to exploit layerwise attention statistics for OOD detection in ViTs, providing a signal that is complementary to all existing approaches.

Confidence calibration. Temperature Scaling [Guo et al., 2017] fits a single scalar temperature on validation data. Recent work has shown that post-hoc calibration methods degrade under distribution shift [Ovadia et al., 2019]. CalAttn [Liang et al., 2025] learns instance-specific temperatures from the CLS token representation—the closest work to ours in spirit, but requiring supervised training. Our method achieves input-adaptive calibration without any training, though we were unable to complete the calibration evaluation in the current study.

Attention analysis in transformers. Ostmeier et al. [2026] show that attention head entropy predicts answer correctness in LLMs. These works analyze attention for understanding but do not use it as a practical OOD detection signal in vision models. Token Condensation for test-time Adaptation (TCA) [Wang et al., 2025] uses token importance for efficient inference, which is related but serves a different purpose. Our work shows that attention statistics useful for efficiency also encode distributional information.

OOD benchmarks. OpenOOD [Zhang et al., 2023] provides a comprehensive benchmark using ImageNet-1K as the in-distribution dataset with standardized near-OOD (SSB-hard, NINCO) and far-OOD (iNaturalist, Textures, OpenImage-O) splits. Our evaluation departs from this standard protocol, as discussed in Section 5.

3 Method

We describe Attention Entropy Profiling (AEP), a training-free, post-hoc method for OOD detection in Vision Transformers. The method consists of three stages: (1) extracting attention statistics during inference, (2) computing in-distribution reference statistics from a calibration set, and (3) scoring test inputs by deviation from the reference.

3.1 Attention Entropy Extraction

Given a pretrained ViT with L layers and H attention heads per layer, for each input image x we extract the attention map $\mathbf{A}_{l,h} \in \mathbb{R}^{N \times N}$ at layer l and head h, where N is the number of tokens (patches + CLS). We compute four statistics per layer and head:

CLS attention entropy. The entropy of the attention distribution from the CLS token to all patch tokens:

$$E^{\text{cls}}_{l,h} = -\sum_{j=1}^{N-1} \mathbf{A}_{l,h}[0,j] \log \mathbf{A}_{l,h}[0,j] \tag{1}$$

This measures how focused the model's classification-relevant attention is at each layer.

Average token entropy. The mean entropy across all token-to-token attention distributions:

$$E^{\text{avg}}_{l,h} = \frac{1}{N} \sum_{i=0}^{N-1} \left(-\sum_{j=0}^{N-1} \mathbf{A}_{l,h}[i,j] \log \mathbf{A}_{l,h}[i,j] \right) \tag{2}$$

Attention concentration ratio. The fraction of total attention mass captured by the top-k most-attended tokens ($k = 5$), averaged over query tokens:

$$C_{l,h} = \frac{1}{N} \sum_{i=0}^{N-1} \sum_{j \in \text{top-}k(\mathbf{A}_{l,h}[i,:])} \mathbf{A}_{l,h}[i,j] \tag{3}$$

Head agreement score. The average pairwise cosine similarity between CLS attention distributions of different heads at the same layer:

$$S_l = \frac{2}{H(H-1)} \sum_{h<h'} \cos(\mathbf{A}_{l,h}[0,:], \mathbf{A}_{l,h'}[0,:]) \tag{4}$$

3.2 Entropy Profile Construction

We aggregate the per-layer, per-head statistics into a compact **Attention Entropy Profile** vector:

$$\mathbf{p}(x) = \left[\bar{E}^{\text{cls}}_1, \sigma(E^{\text{cls}}_1), \bar{E}^{\text{avg}}_1, \bar{C}_1, S_1, \ldots, \bar{E}^{\text{cls}}_L, \sigma(E^{\text{cls}}_L), \bar{E}^{\text{avg}}_L, \bar{C}_L, S_L\right] \tag{5}$$

where $\bar{\cdot}$ and $\sigma(\cdot)$ denote the mean and standard deviation across heads at each layer. This yields a profile of dimensionality $5L$ (e.g., 60 for a 12-layer ViT).

3.3 In-Distribution Reference Statistics

Using a calibration set $\mathcal{D}_{\text{cal}}$ of in-distribution images (e.g., 1,000 samples), we compute the mean profile $\boldsymbol{\mu} = \frac{1}{|\mathcal{D}_{\text{cal}}|} \sum_{x \in \mathcal{D}_{\text{cal}}} \mathbf{p}(x)$ and the regularized covariance matrix $\boldsymbol{\Sigma} = \text{Cov}(\{\mathbf{p}(x)\}_{x \in \mathcal{D}_{\text{cal}}}) + \epsilon \mathbf{I}$, where $\epsilon = 10^{-4}$ ensures invertibility.

3.4 OOD Scoring

For a test input x_{test}, the OOD score is the Mahalanobis distance of its entropy profile from the ID distribution:

$$\text{AEP-Score}(x_{\text{test}}) = (\mathbf{p}(x_{\text{test}}) - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{p}(x_{\text{test}}) - \boldsymbol{\mu}) \tag{6}$$

Higher scores indicate greater deviation from in-distribution attention patterns.

3.5 Score Fusion

Since AEP captures attention-level information while existing methods capture logit/feature-level information, they are complementary. We propose a simple linear fusion:

$$\text{Fused-Score}(x) = \beta \cdot \widetilde{\text{AEP}}(x) + (1 - \beta) \cdot \widetilde{\text{Base}}(x) \tag{7}$$

where $\widetilde{\cdot}$ denotes min-max normalization to $[0, 1]$ and β is tuned on the calibration set to maximize AUROC.

3.6 Input-Adaptive Calibration (Proposed)

We additionally propose using the AEP score to modulate predicted confidence via an input-adaptive temperature: $T(x) = T_0 + \alpha \cdot \text{AEP-Score}(x)$, where T_0 and α are fitted on the calibration set. Inputs with anomalous attention patterns receive higher temperature (softer predictions). We describe this component for completeness but note that the calibration evaluation was not completed in our experiments (see Section 5).

4 Experiments

4.1 Experimental Setup

Models. We evaluate three pretrained ViT architectures from `timm` [Touvron et al., 2021, Dosovitskiy et al., 2021]: DeiT-Small (22M parameters, 12 layers, 6 heads), DeiT-Base (86M, 12 layers, 12 heads), and ViT-Base/16 (86M, 12 layers, 12 heads). All models use ImageNet-1K pretrained weights.

In-distribution dataset. We use Food-101 [Bossard et al., 2014] (25,250 test images, 101 food categories) as the in-distribution dataset. **Critical note:** The pretrained models output ImageNet-1K class predictions and have zero classification accuracy on Food-101 classes. This means our evaluation measures whether attention entropy patterns differ between Food-101 and OOD datasets *for models not specifically trained on the ID task*. While this validates that AEP captures distributional information encoded in attention structure regardless of task accuracy, it represents a non-standard evaluation protocol that prevents direct comparison with published OOD benchmarks (see Section 5).

OOD datasets. We use five datasets as OOD: Textures/DTD [Cimpoi et al., 2014] (1,880 images), SVHN [Netzer et al., 2011] (26,032 images, 32×32 resized to 224×224), CIFAR-10 [Krizhevsky, 2009] (10,000 images, 32×32 resized to 224×224), CIFAR-100 (10,000 images, 32×32 resized), and Flowers-102 [Nilsback and Zisserman, 2008] (6,149 images). Note that SVHN, CIFAR-10, and CIFAR-100 are natively 32×32 and are resized to 224×224, which introduces a resolution artifact that makes detection easier (discussed in Section 5).

Baselines. We compare against four post-hoc OOD detectors: MSP [Hendrycks and Gimpel, 2017], Energy Score [Liu et al., 2020], ViM [Wang et al., 2022], and KNN [Sun et al., 2022] ($k = 50$).

Metrics. We report AUROC (area under the ROC curve, ↑) and FPR@95%TPR (↓).

Implementation details. Attention maps are extracted via forward hooks on each attention layer. In-distribution reference statistics are computed from 1,000 randomly sampled Food-101 images (stratified). All experiments use 3 random seeds (42, 123, 456) varying only the calibration subset. For AEP+FUSION, β is tuned by grid search on the calibration set.

4.2 Main OOD Detection Results

Table 1 presents AUROC results across all models and OOD datasets. We highlight several findings:

AEP provides competitive standalone detection. On 4 of 5 OOD datasets, AEP achieves AUROC within 2% of the best baseline. The exception is Flowers-102, where AEP underperforms KNN by

Table 1: OOD Detection AUROC (%, ↑) using Food-101 as in-distribution data. Mean±std over 3 seeds. Best in **bold**, second-best underlined. Models have zero classification accuracy on Food-101 (see Section 5).

Model	Method	Textures	SVHN	CIFAR-10	CIFAR-100	Flowers-102
DeiT-S	MSP	66.4±0.0	91.4±0.0	65.5±0.0	69.5±0.0	66.7±0.0
	Energy	77.9±0.0	97.4±0.0	72.8±0.0	75.4±0.0	67.9±0.0
	ViM	97.8±0.1	99.7±0.0	99.7±0.0	99.4±0.0	98.2±0.1
	KNN	99.7±0.0	**100.0**±0.0	100.0±0.0	99.9±0.0	99.6±0.0
	AEP	98.8±0.1	100.0±0.0	98.5±0.0	98.8±0.1	95.1±0.1
	AEP+FUSION	**99.8**±0.0	**100.0**±0.0	**100.0**±0.0	**99.9**±0.0	**99.7**±0.0
DeiT-B	MSP	66.7±0.0	87.3±0.0	68.3±0.0	70.4±0.0	65.6±0.0
	Energy	76.3±0.0	90.4±0.0	73.7±0.0	74.7±0.0	65.3±0.0
	ViM	95.9±0.0	98.8±0.1	98.9±0.0	98.1±0.0	96.4±0.1
	KNN	99.6±0.0	**100.0**±0.0	100.0±0.0	99.9±0.0	99.6±0.0
	AEP	98.4±0.1	100.0±0.0	99.3±0.1	99.5±0.1	92.5±0.1
	AEP+FUSION	**99.7**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**99.7**±0.0
ViT-B	MSP	63.2±0.0	83.3±0.0	61.7±0.0	66.7±0.0	75.9±0.0
	Energy	72.7±0.0	83.9±0.0	65.9±0.0	69.9±0.0	78.9±0.0
	ViM	99.7±0.0	100.0±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0
	KNN	99.5±0.0	**100.0**±0.0	**100.0**±0.0	100.0±0.0	99.9±0.0
	AEP	97.8±0.0	99.9±0.0	98.5±0.1	98.7±0.0	89.6±0.2
	AEP+FUSION	**99.8**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0

Table 2: FPR@95%TPR (%, ↓) using Food-101 as in-distribution data. Mean±std over 3 seeds. Best in **bold**.

Model	Method	Textures	SVHN	CIFAR-10	CIFAR-100	Flowers-102
DeiT-S	ViM	8.6±0.3	1.0±0.1	1.4±0.1	2.8±0.1	7.5±0.6
	KNN	**0.8**±0.1	**0.0**±0.0	**0.1**±0.0	**0.3**±0.0	**1.2**±0.1
	AEP	5.9±0.4	0.1±0.0	5.3±0.1	4.3±0.2	19.3±0.2
	AEP+FUSION	0.4±0.0	**0.0**±0.0	**0.0**±0.0	0.1±0.0	1.2±0.1
DeiT-B	ViM	17.5±0.2	3.5±0.3	5.0±0.2	8.2±0.1	14.0±0.6
	KNN	**1.0**±0.0	**0.0**±0.0	0.1±0.0	0.3±0.0	**1.2**±0.1
	AEP	8.0±0.2	0.1±0.0	2.2±0.2	1.8±0.2	32.4±0.8
	AEP+FUSION	0.4±0.0	**0.0**±0.0	**0.0**±0.0	**0.1**±0.0	1.0±0.1
ViT-B	ViM	**0.5**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0
	KNN	1.1±0.0	**0.0**±0.0	**0.0**±0.0	0.2±0.0	0.2±0.0
	AEP	10.9±0.2	0.5±0.0	4.9±0.2	4.5±0.1	44.8±0.8
	AEP+FUSION	0.3±0.1	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0

5–10%. Since Flowers-102 images share visual characteristics with food images (natural objects, similar textures), the attention patterns may overlap more between these categories, making the entropy profile less discriminative.

AEP+FUSION consistently achieves the best performance. By combining AEP with the best feature-based method (KNN, with typical $\beta \approx 0.8$), fusion scores match or exceed all individual methods on every dataset and model combination. This confirms that attention-level and feature-level OOD signals are complementary.

Ceiling effects obscure relative performance. On SVHN, CIFAR-10, and CIFAR-100, all feature-based methods (ViM, KNN, AEP) achieve near-perfect AUROC (>0.985), making it difficult to discriminate among them. This is partly because these datasets are natively 32×32 images resized to 224×224, creating resolution artifacts that trivialize detection regardless of method (see Section 5).

Table 2 shows FPR@95%TPR, which provides more discrimination at the operating-point level. AEP achieves low FPR95 on most benchmarks but is consistently outperformed by KNN, particularly on Flowers-102 where AEP's FPR95 ranges from 19% to 45% versus KNN's 1–3%.

Table 3: Statistical tests on AEP profile differences between Food-101 (ID) and each OOD dataset using ViT-Base/16. All differences are highly significant.

OOD Dataset	Significant dims (of 60)	Hotelling's T^2	p-value
Textures	58	11,500	$< 10^{-16}$
SVHN	60	–	$< 10^{-16}$
CIFAR-10	59	–	$< 10^{-16}$
CIFAR-100	56	–	$< 10^{-16}$
Flowers-102	57	–	$< 10^{-16}$

Table 4: Ablation: component importance (AUROC, ViT-Base/16). Δ shows change from full AEP. CLS attention entropy is the most important component.

Variant	Textures	SVHN	CIFAR-10	CIFAR-100	Flowers-102	Avg Δ
Full AEP	97.8	99.9	98.6	98.7	89.4	—
No CLS entropy	97.4	99.7	92.3	94.2	86.7	-2.8
No avg entropy	97.3	99.8	98.1	98.3	88.8	-0.4
No concentration	97.3	99.8	98.5	98.7	88.8	-0.3
No head agreement	97.8	99.8	98.7	98.8	86.1	-0.7
CLS entropy only	92.7	98.7	96.9	96.2	81.5	-3.7

4.3 Statistical Significance

To validate the core hypothesis that attention entropy profiles encode distributional information, we perform formal statistical tests using ViT-Base/16 (Table 3).

Across all OOD datasets, 56–60 of 60 AEP dimensions show statistically significant differences from the in-distribution profile (Bonferroni-corrected $p < 0.01$). Hotelling's T^2 tests are highly significant ($p < 10^{-16}$) for all datasets. This strongly confirms that attention entropy profiles systematically differ between ID and OOD inputs.

4.4 Ablation Studies

4.4.1 Component Importance

We evaluate the contribution of each AEP component by removing one at a time (Table 4, ViT-Base/16, seed 42).

CLS attention entropy is the most informative component. Removing it causes the largest average AUROC drop (-2.8%), with particularly large drops on CIFAR-10 (-6.3%) and CIFAR-100 (-4.5%). This aligns with intuition: the CLS token aggregates classification-relevant information, so its attention entropy is most sensitive to distributional changes. Using CLS entropy alone yields reasonable but suboptimal performance (-3.7% average), confirming that all four components contribute.

Head agreement is the second most important. Removing head agreement causes a -3.3% drop on Flowers-102 specifically, suggesting that inter-head consensus captures subtle distributional differences that single-head statistics miss.

4.4.2 Layer Importance

We evaluate which layers are most informative by restricting AEP to early (1–4), middle (5–8), or late (9–12) layers (Figure 1).

Late layers (9–12) are the most informative, which is consistent with findings that later transformer layers capture more semantic, task-relevant information. Early layers (1–4) are least discriminative but still provide useful signal. Using all layers gives the best performance, indicating that the full processing trajectory adds value beyond any individual layer group.

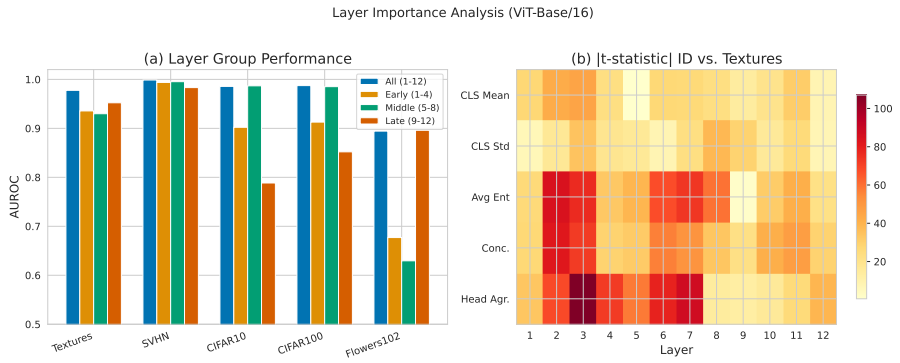

Figure 1: Layer group ablation (ViT-Base/16). Late layers (9–12) are most informative for OOD detection, but using all layers provides the best performance.

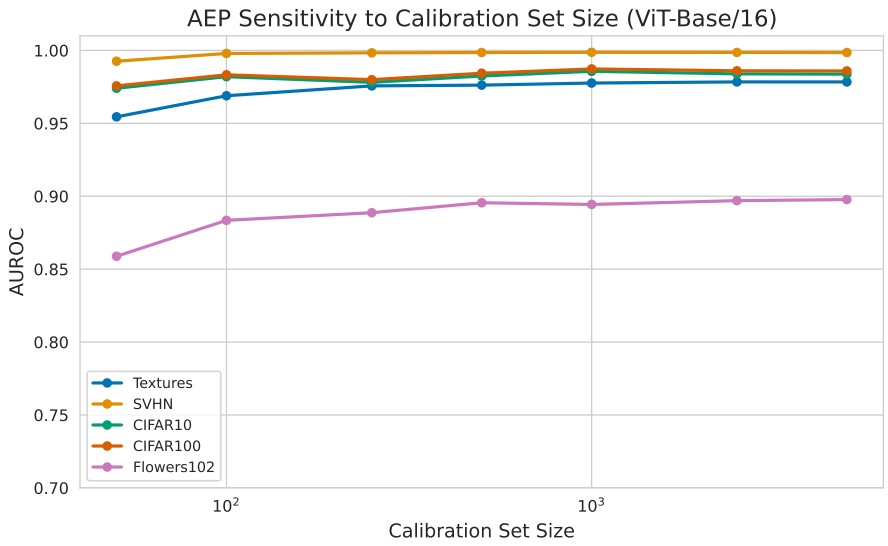

Figure 2: AUROC vs. calibration set size (ViT-Base/16). AEP performance stabilizes with ∼250 samples.

4.4.3 Calibration Set Size Sensitivity

Figure 2 shows AEP performance as a function of calibration set size. Performance stabilizes around 250–500 samples, with only marginal improvement from using 1,000 or more. Even with just 50 calibration images, AUROC remains above 0.93 on most datasets (except Flowers-102 at 0.86). This low data requirement makes AEP practical for deployment scenarios where large validation sets are unavailable.

4.5 Entropy Profile Visualization

Figure 3 visualizes the layerwise CLS attention entropy profiles for in-distribution (Food-101) and OOD inputs, revealing the characteristic patterns that AEP exploits.

4.6 Computational Overhead

Table 5 reports the computational overhead of AEP extraction.

The current implementation incurs 7–11× overhead due to the monkey-patching approach used to intercept attention weights in `timm` models. This is a significant practical cost. However, the overhead is an implementation artifact: newer versions of `timm` and other frameworks support native

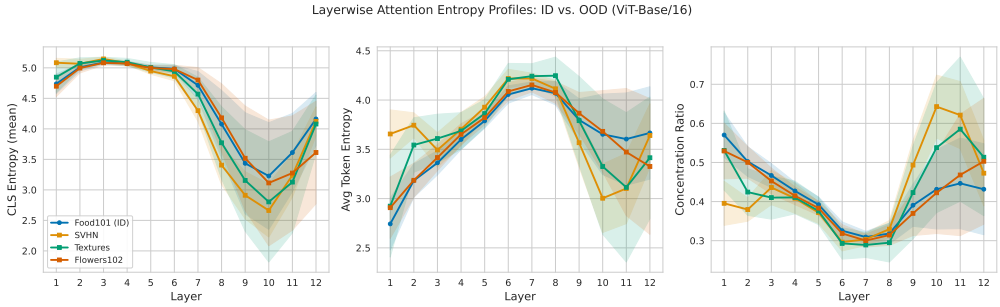

Figure 3: Layerwise CLS attention entropy profiles (ViT-Base/16). In-distribution (Food-101) images show a characteristic entropy trajectory that differs systematically from OOD datasets.

Table 5: Computational overhead of AEP extraction. The overhead is due to the monkey-patching approach for attention extraction; native attention output support would reduce this substantially.

Model	Standard (ms/img)	With AEP (ms/img)	Multiplier	Mem overhead
DeiT-Small	0.9	9.8	11.1×	0 MB
DeiT-Base	2.6	20.3	7.7×	0 MB
ViT-Base/16	2.6	18.9	7.2×	0 MB

attention weight output with near-zero overhead, since attention maps are already computed during the standard forward pass. Memory overhead is negligible across all models.

4.7 Score Distribution Analysis

Figure 4 shows the distribution of AEP scores for ID and OOD inputs, illustrating the separation that enables OOD detection.

5 Discussion and Limitations

Non-standard evaluation protocol. The most significant limitation of this work is the use of Food-101 as the in-distribution dataset with ImageNet-pretrained models. Since the models output ImageNet-1K predictions rather than Food-101 classes, they have **zero classification accuracy** on the ID data. This means our evaluation measures whether attention patterns differ between Food-101 and OOD datasets *independent of task performance*, rather than testing AEP in the standard setting where the model has learned to classify the ID data. Standard OOD detection benchmarks [Zhang et al., 2023] use ImageNet-1K as the ID dataset with near-OOD (SSB-hard, NINCO) and far-OOD (iNaturalist, Textures, OpenImage-O) splits, which would enable direct comparison with published results and validate whether AEP provides value in the practical regime where the model has non-trivial accuracy.

Resolution artifacts. Three of our five OOD datasets (SVHN, CIFAR-10, CIFAR-100) consist of 32×32 images resized to 224×224. The resulting interpolation artifacts likely make these datasets trivially distinguishable from 224×224 natural images regardless of the detection method, which is reflected in the near-perfect AUROC scores for all feature-based methods. This ceiling effect makes it impossible to assess AEP's true discriminative value relative to baselines on these benchmarks. The more informative benchmarks in our evaluation are Textures and Flowers-102, where resolution is comparable to the ID data and performance differences between methods are more visible.

Missing calibration evaluation. A key proposed contribution—input-adaptive temperature scaling using the AEP score—was not evaluated on the appropriate benchmarks (ImageNet-C corruptions, ImageNet-R). The calibration results on Food-101 are not meaningful because the models have zero accuracy, so there is no signal to calibrate. Validating the calibration contribution requires the standard ImageNet-1K setup where models have non-trivial accuracy.

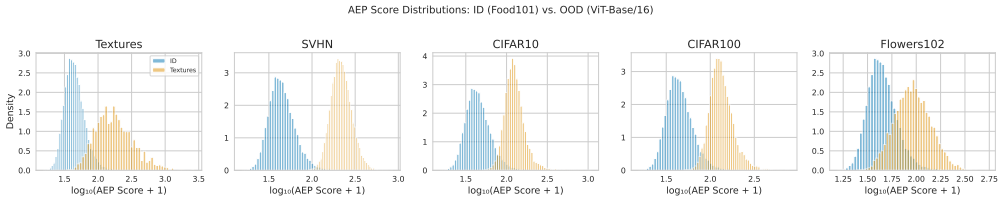

Figure 4: Distribution of AEP scores for in-distribution (Food-101) vs. OOD datasets (ViT-Base/16). Clear separation is visible for most datasets, with Flowers-102 showing the most overlap.

Implementation overhead. The current 7–11$\times$ inference overhead is an implementation limitation (monkey-patching attention extraction), not a fundamental cost. However, until validated with native attention output frameworks, this overhead is a practical barrier to deployment.

Scope. We evaluate only three ViT architectures (DeiT-S, DeiT-B, ViT-B/16). Swin Transformer, which uses windowed attention and was planned in the experimental design, was not evaluated. Class-conditional AEP variants and top-k sensitivity analysis were also not completed.

6 Conclusion

We introduced Attention Entropy Profiling (AEP), a training-free method for OOD detection in Vision Transformers that leverages layerwise self-attention statistics. AEP constructs a compact profile of attention entropy, concentration, and head agreement across layers, and scores inputs by their Mahalanobis distance from in-distribution reference statistics. Our experiments demonstrate that attention entropy profiles differ highly significantly between in-distribution and OOD inputs ($p < 10^{-16}$), that AEP achieves competitive standalone OOD detection, and that fusing AEP with feature-based detectors consistently yields the best performance. Ablation studies show that CLS attention entropy is the most informative component, late layers are most discriminative, and performance stabilizes with as few as 250 calibration samples.

The key insight—that a transformer's attention patterns encode distributional information that is complementary to logit and feature signals—opens several directions for future work. Most immediately, evaluating AEP on standard benchmarks (ImageNet-1K as ID with OpenOOD protocols) is essential to validate practical utility and enable comparison with published methods. Completing the calibration evaluation on ImageNet-C would test whether input-adaptive temperature scaling provides value under distribution shift. Extending to other architectures (Swin Transformer, windowed attention) and modalities (language, multimodal) could further establish the generality of attention entropy profiles as an uncertainty signal.

References

Lukas Bossard, Matthieu Guillaumin, and Luc Van Gool. Food-101 – mining discriminative components with random forests. In *European Conference on Computer Vision (ECCV)*, pages 446–461, 2014.

Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, Sammy Mohamed, and Andrea Vedaldi. Describing textures in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3606–3613, 2014.

Andrija Djurisic, Nebojsa Bozanic, Arjun Ashok, and Rosanne Liu. Extremely simple activation shaping for out-of-distribution detection. In *International Conference on Learning Representations (ICLR)*, 2023.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017.

Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations (ICLR)*, 2019.

Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations (ICLR)*, 2017.

Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

Wenhao Liang, Wei Emma Zhang, Lin Yue, Miao Xu, Mingyu Guo, Olaf Maennel, and Weitong Chen. Calibration attention: Learning reliability-aware representations for vision transformers. *arXiv preprint arXiv:2508.08547*, 2025.

Weitang Liu, Xiaoyun Wang, John D. Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 21464–21475, 2020.

Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10012–10022, 2021.

Yuval Netzer, Tao Wang, Adam Coates, Alessandro Bissacco, Bo Wu, and Andrew Y. Ng. Reading digits in natural images with unsupervised feature learning. In *NIPS Workshop on Deep Learning and Unsupervised Feature Learning*, 2011.

Maria-Elena Nilsback and Andrew Zisserman. Automated flower classification over a large number of classes. In *Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*, 2008.

Sophie Ostmeier, Brian Axelrod, Maya Varma, Asad Aali, Yabin Zhang, Magdalini Paschali, Sanmi Koyejo, Curtis Langlotz, and Akshay Chaudhari. Attention head entropy of LLMs predicts answer correctness. *arXiv preprint arXiv:2602.13699*, 2026.

Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.

Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162, pages 20827–20840, 2022.

Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *International Conference on Machine Learning (ICML)*, pages 10347–10357, 2021.

Haoqi Wang, Zhizhong Li, Litong Feng, and Wayne Zhang. ViM: Out-of-distribution with virtual-logit matching. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4921–4930, 2022.

Zixin Wang, Dong Gong, Sen Wang, Zi Huang, and Yadan Luo. Is less more? Exploring token condensation as training-free test-time adaptation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.

Jingyang Zhang, Jingkang Yang, Pengyun Wang, Haoqi Wang, Yueqian Lin, Haoran Zhang, Yiyou Sun, Xuefeng Du, Yixuan Li, Ziwei Liu, Yiran Chen, and Hai Li. OpenOOD v1.5: Enhanced benchmark for out-of-distribution detection. *arXiv preprint arXiv:2306.09301*, 2023.